

Gesture control

The gesture control section consists of transmitter and receiver units.

Transmitter unit. The circuit diagram of the gesture transmitter unit

is shown in Fig. 2. It is built around Atmega328 MCU chip (IC9), ADXL335 accelerometer (IC8), 7805 voltage regulator (IC8), LM1117-3.3 voltage regulator (IC10), HT12E encoder (IC11), 433MHz RF transmitter module (RF TX1) and a

few other components.

ADXL335 accelerometer. Normally, an accelerometer denotes positive x, y and z axes on the sensor package. These are defined so that linear acceleration aligned in the direction of

these axes will give a positive accelerometer output. ADXL335 gives complete three-axis acceleration measurement within the range $\pm 3g$ in x, y and z axes. Output signals of this module are analogue voltages that are proportional to acceleration.

The transmitter along with ADXL335 sensor can be attached to your hand or palm. When you move your hand along with the sensor, the positional or gesture signal is processed by Atmega328 MCU and transmitted to the receiver unit through encoder IC11 and RF TX1 module. That is, by using different hand gestures, you can move the robot in different directions.

Arduino programming. Arduino programming tool is used to program ATmega328 MCU. The software program is written in Arduino programming language. You can program a fresh ATmega328 MCU with the help of Arduino IDE and Arduino Uno board.

First, load bootloader code into the MCU. For that, use Arduino

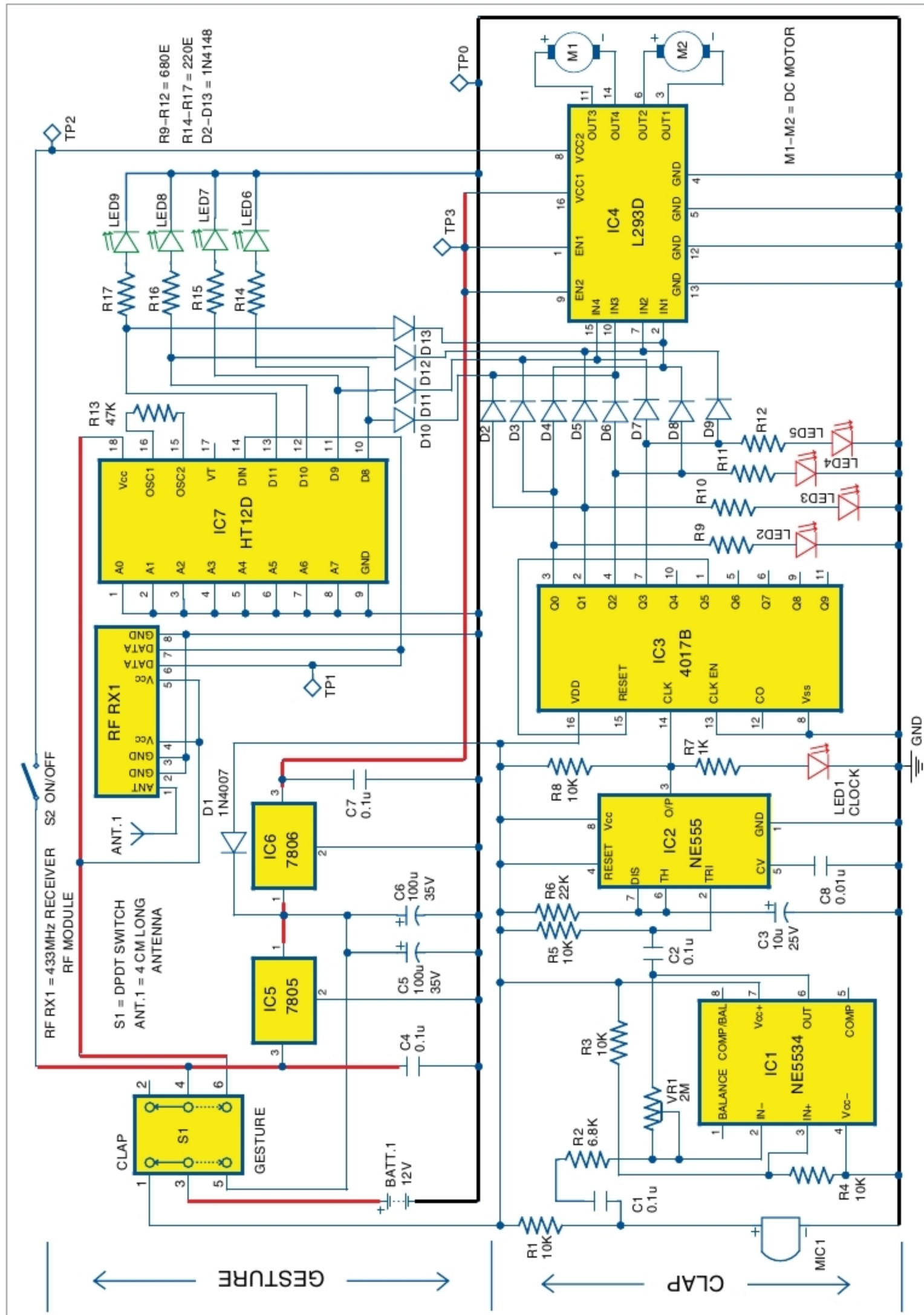


Fig. 1: Circuit diagram of the clap- and gesture-controlled robot (Receiver)